

Product Insert

Enrovet, 100mg/ml, Solution for use in drinking water

Active ingredient:

Each ml contains
Enrofloxacin-----100mg

Preservative:

Each ml contains
Sodium benzoate-----2mg

Product description:

Yellowish colour liquid

Dosage form:

Solution for use in drinking water

Pharmacodynamics:

Mode of action-

Two enzymes essential in DNA replication and transcription, DNA gyrase and topoisomerase IV, have been identified as the molecular targets of fluoroquinolones. They modulate the topological state of DNA through cleaving and resealing reactions. Initially, both strands of the DNA double helix are cleaved. Then, a distant segment of DNA is passed through this break before the strands are resealed. Target inhibition is caused by non-covalent binding of fluoroquinolone molecules to an intermediate state in this sequence of reactions, in which DNA is cleaved, but both strands are retained covalently attached to the enzymes. Replication forks and translational complexes cannot proceed beyond such enzyme-DNA-fluoroquinolone complexes, and inhibition of DNA and mRNA synthesis triggers events resulting in a rapid, drug concentration-dependant killing of pathogenic bacteria.

Antibacterial spectrum

Enrofloxacin is active against many Gram-negative bacteria, against Gram-positive bacteria and *Mycoplasma* spp.

In vitro susceptibility has been shown in strains of (i) Gram-negative species such as, *Pasteurella multocida* and *Avibacterium (Haemophilus) paragallinarum* and (ii) *Mycoplasma gallisepticum* and *Mycoplasma synoviae*.

Types and mechanisms of resistance

Resistance to fluoroquinolones has been reported to arise from five sources, (i) point mutations in the genes encoding for DNA gyrase and/or topoisomerase IV leading to alterations of the respective enzyme, (ii) alterations of drug permeability in Gram-negative bacteria, (iii) efflux mechanisms, (iv) plasmid mediated resistance and (v) gyrase protecting proteins. All mechanisms lead to a reduced susceptibility of the bacteria to fluoroquinolones. Cross-resistance within the fluoroquinolone class of antimicrobials is common.

Pharmacokinetics:

Enrofloxacin administered via drinking water to poultry is rapidly and very well absorbed with a bioavailability of approximately 90%. Maximum plasma concentrations of 2 mg/L are reached within 1.5 hours after a single bolus dose rate of 10 mg/kg body weight with a total systemic availability of 14.4 mg hr/L. Enrofloxacin is eliminated from the body with a total body clearance of 10.3 ml/min kg. If dosed as continuous drinking water medication (multiple dosing) steady-state concentrations of 0.5 mg (turkeys) to 0.8 mg (chicken) enrofloxacin per litre are achieved. A high mean volume of distribution (5 L/kg) indicated good tissue penetration of enrofloxacin. Concentrations in target tissues like lungs, liver, kidney, intestine and muscle tissue, exceed plasma concentrations by far. In poultry enrofloxacin is poorly metabolised to its active metabolite ciprofloxacin (approximately 5%). Enrofloxacin is eliminated from the body at a half-life of 6 hours. Protein binding in poultry is approximately 25%.

Indication:

Treatment of infections caused by the following bacteria susceptible to enrofloxacin:

Chickens- *Mycoplasma gallisepticum*, *Mycoplasma synoviae*, *Avibacterium paragallinarum*, *Pasteurella multocida*;

Turkey- *Mycoplasma gallisepticum*, *Mycoplasma synoviae*, *Pasteurella multocida*;

Rabbits- For the treatment infectious diseases due to *Pasteurella multocida* and bacterial enteritis due to infection with *E.coli*.

Enrofloxacin should be used where clinical experience, supported where possible by sensitivity testing of the causal organism, indicates enrofloxacin as the active substance of choice.

Mode of administration:

Oral administration via drinking water

Recommended dosage:

Chickens and turkeys- 10 mg enrofloxacin/kg bodyweight per day for 3-5 consecutive days. Treatment for 3-5 consecutive days; for 5 consecutive days in mixed infections and chronic progressive forms. If no clinical improvement is achieved within 2-3 days, alternative antimicrobial therapy should be considered based on susceptibility testing. Via the drinking water. Always make sure that the entire dose offered has been consumed. The medicated water should be made up fresh each day just before it is offered to the animals.

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The drinking water must be medicated throughout the treatment period, and no other water source should be available. Determine the bodyweight of the birds as accurately as possible in order to avoid underdosing. Use only fresh pre-solutions, prepare every day before start of treatment.

Pumping systems should be checked constantly to assure proper medication. Empty the water system and fill it with medicated water before starting the treatment. Calculate the daily quantity (ml) of product required for treatment period as follows:

Total number of birds x Average body weight in kg x 0.1 = Total volume (ml) per day

The product may be put directly into the header tank or introduced via a water proportioner pump.

Rabbits- 10 mg/kg bodyweight per day for 5 consecutive days. Calculate the daily quantity (ml) of product required for treatment period as follows:

Total number of rabbits x Average body weight in kg x 0.1 = Total volume (ml) per day

(TO BE PRESCRIBED BY REGISTERED VETERINARY SURGEONS ONLY)

Contraindications:

Do not use for prophylaxis. Do not use when resistance/ cross-resistance to (fluoro)quinolones is known to occur in the flock intended for treatment. Do not use in the case of known hypersensitivity to the active substance, other (fluoro)quinolones or to any of the excipients.

Special warnings for each target species:

Treatment of *Mycoplasma spp* infections may not eradicate the organism.

Special precautions for use in animals:

Official and local antimicrobial policies should be taken into account when the product is used.

Fluoroquinolones should be reserved for the treatment of clinical conditions which have responded poorly, or are expected to respond poorly, to other classes of antimicrobials.

Since enrofloxacin was first authorised for use in poultry, there has been widespread reduction in susceptibility of *E.coli* to fluoroquinolones and emergence of resistant organisms. Resistance has also been reported in *Mycoplasma synovia* in the EU.

Whenever possible, fluoroquinolones should only be used based on susceptibility testing.

Use of the product deviating from the instructions given may increase the prevalence of bacteria resistant to the fluoroquinolones and may decrease the effectiveness of treatment with other quinolones due to the potential for cross resistance.

Special precautions to be taken by the person administering the medicinal product to animals:

Those with known hypersensitivity to (fluoro)quinolones should avoid contact with this product. Avoid contact with skin and eyes.

Rinse any splashes from skin or eyes immediately with water. Wash hands and exposed skin after use. Do not eat, drink or smoke while using the product.

Interactions with other medicaments:

In vitro, an antagonism was shown, when combining fluoroquinolones with bacteriostatic antimicrobial agents such as macrolides or tetracyclines and phenicols. The simultaneous application of substances containing aluminium or magnesium can impair the absorption of enrofloxacin.

Usage during pregnancy and lactation:

Do not use in laying hens producing eggs for human consumption. Do not administer to layer replacement birds within 14 days of coming into lay.

Adverse effects:

None known.

Overdose and treatment:

No adverse clinical symptoms were observed in chickens and turkeys treated respectively with doses up to 10 and 6 times higher than the therapy dose.

The use of fluoroquinolones during the growth phase combined with a marked and prolonged increase in the intake of drinking water, and hence active ingredient, possibly due to high temperatures, may potentially be associated with damage of the articular cartilage.

Withdrawal periods:

Chickens: Meat and offal: 7 days.

Turkeys: Meat and offal: 13 days.

Rabbits: Meat and offal: 3 days.

Not authorised for use in birds producing eggs for human consumption.

Do not administer to layer replacement birds within 14 days of coming into lay.

Storage condition:

Protect from direct sunlight. Store in cool, dry place (<30°C).

Shelf life:

24 months

Shelf life after opening:

Discard any unused solution immediately after opening

Shelf life after dilution/reconstitution:

24 hours

Packaging available:

1L/bottle; 12 bottles/carton

Manufacturer:

N.S. Laboratory Sdn. Bhd.

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Registration holder:

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